Abhinav Shivkumar Ovhal

AI & Machine Learning Undergraduate | Full-Stack Developer
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PROFESSIONAL SUMMARY

Artificial Intelligence and Machine Learning undergraduate with hands-on experience in full-stack web development and applied deep learning. Experienced in developing scalable web applications using Next.js, TypeScript, and MongoDB, and building machine learning models using TensorFlow and Keras. Strong foundation in data structures, problem-solving, and end-to-end project implementation.

EDUCATION

National Institute of Technology, Kurukshetra Aug 2023 – June 2027
B.Tech in Artificial Intelligence and Machine Learning

Kendriya Vidyalaya No.1 AFS Pune CBSE Class XII | 89.4%

PROJECTS

Ethereum Phishing Account Detection live link
Python, PyTorch, Streamlit, Scikit-learn, Etherscan API

- **Problem:** Standard models fail on Ethereum transaction graphs due to extreme class imbalance (1.5% phishing) and temporal data leakage.
- **Solution:** Engineered a PyTorch cross-stream attention fusion layer combining two independent GNN pipelines (DA-HGNN & GAE+AdaBoost).
- **Features:** Built a dynamic mechanism weighting latent embeddings against raw probabilities, and eliminated data leakage via strict chronological node splits from 44,000+ edge timestamps.
- **Impact:** Boosted F1-score by **7.2x** and achieved 0.850 AUC-ROC over baselines, proving robust performance at a 40:1 imbalance ratio.

OpenClaw (Autonomous AI Agent) GitHub
TypeScript, Bun, SQLite, Vercel AI SDK, OpenRouter

- **Problem:** Standard AI agents corrupt workspaces, fail at edits due to LLM whitespace hallucinations, and suffer from multi-agent race conditions.
- **Solution:** Engineered a concurrent AI platform utilizing a Virtual File Overlay to guarantee 100% safe, zero-disk-mutation code refactoring.
- **Features:** Architected a 3-tier cascading fuzzy search algorithm for resilient edits, lock-free SQLite WAL task queues, and a hybrid AST/Vector semantic indexer.
- **Impact:** Cut LLM token costs by **∼80%**, eliminated multi-worker race conditions, and secured the execution pipeline against shell injection attacks.

AI vs Real Image Detection System GitHub
React (TypeScript), Flask, TensorFlow, Keras, CNN

- **Problem:** The rapid advancement of generative AI makes it increasingly difficult to visually distinguish deepfakes and AI-generated imagery from real photographs.
- **Solution:** Developed a full-stack deep learning platform that provides real-time, drag-and-drop binary classification of uploaded images.
- **Features:** Implemented a custom Inception-based CNN with an automated preprocessing pipeline (224x224 resizing, normalization, RGB conversion) integrated via a RESTful Flask API.
- **Impact:** Delivered a highly responsive, low-latency web tool that accurately identifies modern AI-generated content, empowering users with instant visual feedback and confidence scores.

SKILLS

- **Hard Skills:** Machine Learning, Deep Learning, CNN, LSTM, REST API Development, Full-Stack Web Development, Database Design
- **Programming Languages:** JavaScript, TypeScript, Python, C, C++
- **Frameworks & Tools:** React.js, Next.js, Node.js, Express.js, TensorFlow, Keras, Pandas, NumPy, MongoDB, PostgreSQL, SQL, LangChain, N8N
- **Soft Skills:** Problem Solving, Analytical Thinking, Attention to Detail, Self-Learning, Time Management, Team Collaboration

ACHIEVEMENTS AND CERTIFICATIONS

- Solved **500+ DSA problems** on various platforms Codolio
- Completed **Machine Learning Specialization** – Regression & Classification Coursera
- Completed **Nodejs- Begineer to Advance course** Certificate
- Writes software related articles on **Medium** – Medium